Lab Assignment 2

Stepan Shkrobak

Dr. Michael Bartolacci

IST 894, Section 001: Capstone Experience,

8 February 2026

1. General Context....2
2. Technical Context....3
3. Resources....5
4. Screenshots....6
Nmap and Hping....6
Applied Nmap....7
Basic Scripting....9
Lab Completion....10

1. General Context

Network reconnaissance is key to any attack or defense development. In these labs, various tools and methods are used to discover networks, enabling the identification of potential vulnerabilities that can then be used for further and deeper reconnaissance or even attack. Nmap is one of the tools that allows for identifying active hosts, determining open network ports, and identifying service and operating system information without actually exploiting them. This is similar to a property inspector walking through a building to note every door, window, and entry point before deciding where improvements are needed. Discovering hosts and software versions is one of the simplest ways to find vulnerabilities in known software versions that can be exploited.

In the applied Nmap lab, we perform an Nmap scan, which reveals an FTP service that is running with an outdated software version and weak configuration. Legitimate credentials are then used against the FTP service, granting us access, confirming the vulnerability, and authenticating and accessing the files. This workflow mirrors what penetration testers do in real engagements: find the weak door, check if

the lock works, and then determine whether known flaws can be exploited. This access could allow an attacker to further exploit the vulnerability and move across the network laterally or load new data onto the FTP in addition to accessing and being able to take data off.

The labs finish off with custom scripting in Python and Bash. Scripting allows for automating repetitive tasks, for example, predefining all the commands we normally would run in the terminal into one script file, which, when run, executes all the commands one after the other at a faster rate than a human typing could do. These skills enable security professionals to build tailored tools for repetitive tasks or integrate multiple utilities into coherent workflows. This is great for organizations as they can develop complex scripts that they can run to perform a complete test of their network, and have it saved to repeat, for example, after they make patches to see if the vulnerability that was identified is no longer there, or now have the script to be able to run on a scheduled basis.

2. Technical Context

Network reconnaissance forms the technical baseline for both offensive security testing and defensive risk assessment because it provides measurable visibility into exposed network services and system characteristics. Using tools such as Nmap, we enumerate live hosts, identify listening TCP and UDP ports, and fingerprint services and operating systems through packet analysis and response behavior. These scans rely on techniques such as SYN probing, banner grabbing, and stack fingerprinting rather than exploitation, which allows us to map the attack surface without altering target systems.

The resulting data set, including port states and detected software versions, can then be correlated with known vulnerability databases to identify systems that may be running unpatched or insecure services.

In the applied Nmap portion of the lab, reconnaissance advanced into service enumeration and exposure validation. Version detection identified an FTP daemon running an outdated release with insecure configuration indicators, suggesting susceptibility to known weaknesses or poor access control. After confirming the service was reachable, authenticated access using valid credentials demonstrated that the exposure translated into functional system access. This step verified that the risk was not theoretical but operational, since authenticated FTP access enables file retrieval, file upload, and potential staging of additional tools or payloads. In real network environments, this type of foothold can support privilege escalation attempts, data exfiltration, or lateral movement through credential reuse and trust relationships.

The scripting component of the lab reinforced the technical mechanics behind automation in security operations. Using Python socket programming, we programmatically established TCP connections and interpreted connection states to determine port accessibility, effectively recreating simplified scanning logic at the code level. In parallel, Bash scripting demonstrated how command execution, exit codes, and conditional logic can be combined to automate multi-step testing workflows. These techniques enable repeatable and scalable assessments, reduce manual execution errors, and support continuous validation by allowing the same scripted tests to be rerun after system updates or configuration changes. Results could be output to various file formats so that they can be ingested for complex analysis of all the outputs and results.

3. Resources

Nmap Project. (n.d.). Nmap: Network mapper. https://nmap.org/

Kumar, A. (2023). Exploiting FTP vulnerabilities for effective penetration testing. Medium. https://medium.com/%401200km/exploiting-ftp-vulnerabilities-for-effective-penetration-testing-a2810df78602

SecurityScorecard. (n.d.). FTP security risks organizations should know about. https://securityscorecard.com/blog/ftp-security-risks

OWASP Foundation. (n.d.). OWASP: Open Web Application Security Project. https://owasp.org/

4. Screenshots

Nmap and Hping

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:32 EST
Nmap scan report for target (192.168.1.101)
Host is up (0.000067s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
25/tcp   open  smtp
3306/tcp open  mysql
3389/tcp open  ms-wbt-server
MAC Address: 00:0C:29:76:92:87 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds
root@kali:/home/kali# cat normal_output
# Nmap 7.94SVN scan initiated Sun Feb  8 12:32:42 2026 as: nmap -oN normal_output target
Nmap scan report for target (192.168.1.101)
Host is up (0.000067s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
25/tcp   open  smtp
3306/tcp open  mysql
3389/tcp open  ms-wbt-server
MAC Address: 00:0C:29:76:92:87 (VMware)

# Nmap done at Sun Feb  8 12:32:42 2026 -- 1 IP address (1 host up) scanned in 0.18 seconds
root@kali:/home/kali# nmap target -oX xml_format.xml
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:32 EST
Nmap scan report for target (192.168.1.101)
Host is up (0.000056s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
25/tcp   open  smtp
3306/tcp open  mysql
3389/tcp open  ms-wbt-server
MAC Address: 00:0C:29:76:92:87 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
root@kali:/home/kali# cat xml_format.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/bin/../share/nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.94SVN scan initiated Sun Feb  8 12:32:47 2026 as: nmap -oX xml_format.xml target -->
<nmaprun scanner="nmap" args="nmap -oX xml_format.xml target" start="1770571967" startstr="Sun Feb  8 12:32:47 2026" versio
n="7.94SVN" xmloutputversion="1.05">
<scaninfo type="syn" protocol="tcp" numservices="1000" services="1,3-4,6-7,9,13,17,19-26,30,32-33,37,42-43,49,53,70,79-85,8
8-90,99-100,106,109-111,113,119,125,135,139,143-144,146,161,163,179,199,211-212,222,254-256,259,264,280,301,306,311,340,366
,389,406-407,416-417,425,427,443-445,458,464-465,481,497,500,512-515,524,541,543-545,548,554-555,563,587,593,616-617,625,63
1,636,646,648,666-668,683,687,691,700,705,711,714,720,722,726,749,765,777,783,787,800-801,808,843,873,880,888,898,900-903,9
11-912,981,987,990,992-993,995,999-1002,1007,1009-1011,1021-1100,1102,1104-1108,1110-1114,1117,1119,1121-1124,1126,1130-113
2,1137-1138,1141,1145,1147-1149,1151-1152,1154,1163-1166,1169,1174-1175,1183,1185-1187,1192,1198-1199,1201,1213,1216-1218,1
233-1234,1236,1244,1247-1248,1259,1271-1272,1277,1287,1296,1300-1301,1309-1311,1322,1328,1334,1352,1417,1433-1434,1443,1455
```

Service Version Detection

```
55055-55056,55555,55600,56737-56738,57294,57797,58080,60020,60443,61532,61900,62078,63331,64623,64680,65000,65129,65389"/>
</extraports>
<port protocol="tcp" portid="21"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="ftp" method="table" conf="3"/></port>
<port protocol="tcp" portid="22"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="ssh" method="table" conf="3"/></port>
<port protocol="tcp" portid="25"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="smtp" method="table" conf="3"/></port
>
<port protocol="tcp" portid="3306"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="mysql" method="table" conf="3"/></p
ort>
<port protocol="tcp" portid="3389"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="ms-wbt-server" method="table" conf=
"3"/></port>
</ports>
<os><portused state="open" proto="tcp" portid="21"/>
<portused state="closed" proto="tcp" portid="1"/>
<portused state="closed" proto="udp" portid="40540"/>
<osmatch name="Linux 4.15 - 5.8" accuracy="100" line="67250">
<osclass type="general purpose" vendor="Linux" osfamily="Linux" osgen="4.X" accuracy="100"><cpe>cpe:/o:linux:linux_kernel:4</cpe></osclass>
<osclass type="general purpose" vendor="Linux" osfamily="Linux" osgen="5.X" accuracy="100"><cpe>cpe:/o:linux:linux_kernel:5</cpe></osclass>
</osmatch>
</os>
<uptime seconds="1011161" lastboot="Tue Jan 27 19:42:36 2026"/>
<distance value="1"/>
<tcpsequence index="260" difficulty="Good luck!" values="B66934CA,B23F525B,A9969297,E5B5D1BB,87F86884,5982A32E"/>
<ipidsequence class="All zeros" values="0,0,0,0,0,0"/>
<tcptssequence class="1000HZ" values="3C4510E5,3C451149,3C4511AE,3C451212,3C451276,3C4512DA"/>
<times srtt="87" rttvar="6" to="100000"/>
</host>
<runstats><finished time="1770572117" timestr="Sun Feb  8 12:35:17 2026" summary="Nmap done at Sun Feb  8 12:35:17 2026; 1 IP address (1 ho
st up) scanned in 1.54 seconds" elapsed="1.54" exit="success"/><hosts up="1" down="0" total="1"/>
</runstats>
</nmaprun>
root@kali:/home/kali# nmap -sV target -oX version_scan.xml
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:35 EST
Nmap scan report for target (192.168.1.101)
Host is up (0.000065s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
21/tcp   open  ftp           vsftpd 2.3.4
22/tcp   open  ssh           OpenSSH 9.4p1 Debian 1 (protocol 2.0)
25/tcp   open  smtp          Postfix smtpd
3306/tcp open  mysql?
3389/tcp open  ms-wbt-server xrdp
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org
/cgi-bin/submit.cgi?new-service :
SF-Port3306-TCP:V=7.94SVN%I=7%D=2/8%Time=6988C96E%P=x86_64-pc-linux-gnu%r(
SF:NULL,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20a
SF:llowed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server")%r(GetReque
SF:st,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20all
SF:owed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server")%r(HTTPOption
SF:s,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20allo
SF:wed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server");
MAC Address: 00:0C:29:76:92:87 (VMware)
Service Info: Host:  kali; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.59 seconds
root@kali:/home/kali#
```

Service Version Detection

Applied Nmap

```
kali@kali:~$ nmap -sV target
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:49 EST
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 80.00% done; ETC: 12:49 (0:00:02 remaining)
Nmap scan report for target (192.168.1.101)
Host is up (0.00014s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT     STATE SERVICE       VERSION
21/tcp   open  ftp           vsftpd 2.3.4
22/tcp   open  ssh           OpenSSH 9.4p1 Debian 1 (protocol 2.0)
25/tcp   open  smtp          Postfix smtpd
3306/tcp open  mysql?
3389/tcp open  ms-wbt-server xrdp
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at
https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3306-TCP:V=7.94SVN%I=7%D=2/8%Time=6988CCB1%P=x86_64-pc-linux-gnu%r(
SF:NULL,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20a
SF:llowed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server")%r(GenericL
SF:ines,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20a
SF:llowed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server")%r(HTTPOpti
SF:ons,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20al
SF:lowed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server")%r(SMBProgNe
SF:g,4C,"H\0\0\x01\xffj\x04Host\x20'192\.168\.1\.100'\x20is\x20not\x20allo
SF:wed\x20to\x20connect\x20to\x20this\x20MariaDB\x20server");
Service Info: Host:  kali; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.29 seconds
kali@kali:~$
```

Scanning the Target

```
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
kali@kali:~$ nmap target -p 21 -sV -sC
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:51 EST
Nmap scan report for target (192.168.1.101)
Host is up (0.00020s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.1.100
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 4
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.51 seconds
kali@kali:~$ nc target 21
220 (vsFTPd 2.3.4)
user anonymous
331 Please specify the password.
pass anonymous
230 Login successful.
```

Leveraging Open Ports for Social Engineering and Credential Attacks

```
OraclMySQL 5.1.48 - 'LOAD DATA INFILE' Denial of Service                          | linux/dos/34510.txt
paBugs 2.0 Beta 3 - 'class.mysql.php' Remote File Inclusion                       | php/webapps/2437.php
PHP 4.4.0 - 'mysql_connect function' Local Buffer Overflow                        | windows/local/1406.php
PHP 4.4.7/5.2.3 - MySQL/MySQLi 'Safe_Mode' Bypass                                 | multiple/local/4392.txt
PHP 4.x/5.x MySQL Library - 'Safe_mode' Filesystem Circumvention (1)              | php/remote/21264.php
PHP 4.x/5.x MySQL Library - 'Safe_mode' Filesystem Circumvention (2)              | php/remote/21265.php
PHP 4.x/5.x MySQL Library - 'Safe_mode' Filesystem Circumvention (3)              | php/remote/21266.php
PHP-MySQL-Quiz - SQL Injection                                                    | php/webapps/10876.txt
PHP-Nuke 8.0 Final - 'INSERT' Blind SQL Injection (MySQL)                         | php/webapps/3344.pl
PHP-Nuke Module Eve-Nuke 0.1 - 'mysql.php' Remote File Inclusion                  | php/webapps/3591.txt
PHP/Mysql Site Builder 0.0.2 - 'htm2PHP.php' File Disclosure                      | php/webapps/2969.txt
PHPwebnews 0.2 MySQL Edition - 'det' SQL Injection                                | php/webapps/5999.txt
PHPwebnews 0.2 MySQL Edition - 'id_kat' SQL Injection                             | php/webapps/5998.txt
PHPwebnews 0.2 MySQL Edition - 'SQL' Insecure Cookie Handling                     | php/webapps/6136.txt
Plixer Scrutinizer NetFlow and sFlow Analyzer 9 - Default MySQL Credential (Metasploit)  | windows/remote/20355.rb
Point of Sales (POS) in VB.Net MySQL Database 1.0 - SQL Injection                 | php/webapps/45721.txt
ProFTPd - 'mod_mysql' Authentication Bypass                                       | multiple/remote/8037.txt
PromoteWeb MySQL - 'id' SQL Injection                                             | php/webapps/6577.txt
Robert Zimmerman PHP / MySQL Scripts - Authentication Bypass                      | php/webapps/10260.txt
Symantec Web Gateway 5.0.3.18 - Blind SQL Injection Backdoor via MySQL Triggers   | php/webapps/20044.txt
waldronmatt FullCalendar-BS4-PHP-MySQL-JSON 1.21 - 'description' Cross-Site Scripting  | php/webapps/47548.txt
waldronmatt FullCalendar-BS4-PHP-MySQL-JSON 1.21 - 'start' SQL Injection          | php/webapps/47546.txt
Woltlab Burning Board 2.3.4 - 'Class_DB_MySQL.php' Cross-Site Scripting           | php/webapps/27444.txt
WordPress Plugin Google Document Embedder 2.5.16 - 'mysql_real_escpae_string' Bypass SQL | php/webapps/35447.txt
Zabbix Agent 3.0.1 - 'mysql.size' Shell Command Injection                         | linux/local/39769.txt
Zend Framework 1.11.4 - 'PDO_MySql' Security Bypass                               | linux/remote/35784.php
ZoomStats 1.0.2 - 'mysql.php' Remote File Inclusion                               | php/webapps/2420.txt
---------------------------------------------------------------------------------------------------------------
Shellcodes: No Results
kali@kali:~$ nmap -p 21 -sV target
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-08 12:52 EST
Nmap scan report for target (192.168.1.101)
Host is up (0.00020s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.3.4
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds
kali@kali:~$
```

Leveraging Frameworks and Databases for Exploit Discovery and Testing

Basic Scripting

```
kali@kali: ~                                                    12:20 PM
Click to start dragging "kali@kali: ~"
File  Actions  Edit  View  Help
.230 80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
kali@kali:~$ sudo apt install cisco-auditing-tool
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
The following NEW packages will be installed:
  cisco-auditing-tool
0 upgraded, 1 newly installed, 0 to remove and 931 not upgraded.
Need to get 51.5 kB of archives.
After this operation, 272 kB of additional disk space will be used.
Get:1 http://kali.download/kali kali-rolling/main amd64 cisco-auditing-tool all 1.0-1kali5 [51.5 kB]
Fetched 51.5 kB in 0s (210 kB/s)
Selecting previously unselected package cisco-auditing-tool.
(Reading database ... 366002 files and directories currently installed.)
Preparing to unpack .../cisco-auditing-tool_1.0-1kali5_all.deb ...
Unpacking cisco-auditing-tool (1.0-1kali5) ...
Setting up cisco-auditing-tool (1.0-1kali5) ...
Processing triggers for kali-menu (2023.4.5) ...
kali@kali:~$ CAT PORT_SCAN.PY

Cisco Auditing Tool - g0ne [null0]
Usage:
        -h hostname     (for scanning single hosts)
        -f hostfile     (for scanning multiple hosts)
        -p port #       (default port is 23)
        -w wordlist     (wordlist for community name guessing)
        -a passlist     (wordlist for password guessing)
        -i [ioshist]    (Check for IOS History bug)
        -l logfile      (file to log to, default screen)
        -q quiet mode   (no screen output)

kali@kali:~$ python3 port_scan.py target 20 4000
Initiated scan on host target
Port 21    Open
Port 22    Open
Port 25    Open
Port 3306    Open
Port 3389    Open
Scan finished. 5 out of 3980 ports open
kali@kali:~$
```

Utilizing python for scaning

```
kali@kali:~$ touch ping.sh
kali@kali:~$ ping target -c 1
PING target (192.168.1.101) 56(84) bytes of data.
64 bytes from target (192.168.1.101): icmp_seq=1 ttl=64 time=0.323 ms

--- target ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.323/0.323/0.323/0.000 ms
kali@kali:~$ echo "ping target -c 1" > ping.sh
kali@kali:~$ cat ping.sh
ping target -c 1
kali@kali:~$ ./ping.sh
bash: ./ping.sh: Permission denied
kali@kali:~$ chmod 744 ping.sh
kali@kali:~$ ./ping.sh target
PING target (192.168.1.101) 56(84) bytes of data.
64 bytes from target (192.168.1.101): icmp_seq=1 ttl=64 time=0.189 ms

--- target ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.189/0.189/0.189/0.000 ms
kali@kali:~$
```

```
--- target ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.189/0.189/0.189/0.000 ms
kali@kali:~$ nano ping.sh
kali@kali:~$ ./ping.sh target
PING target (192.168.1.101) 56(84) bytes of data.
64 bytes from target (192.168.1.101): icmp_seq=1 ttl=64 time=0.197 ms

--- target ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.197/0.197/0.197/0.000 ms
kali@kali:~$ nano ping.sh
```

Enhancing a Ping Script with Shebang and Parameters

Lab Completion

Nmap and Hping
Lab · 30 minutes
Learn how to scan for open ports, services and vulnerabilities using these two staples of penetration testing.

Applied Nmap
Lab · 1 hour
Learn how to apply Nmap to real-world tasks.

Basic Scripting
Lab · 1 hour
Learn some basic Python, Bash and Ruby